Multiple-Instance Metric Learning Network for Hyperspectral Target Detection

Bo Yang, Yi He, Changzhe Jiao, *Member, IEEE*, Xiao Pan, Guozhen Wang, Lei Wang, and Jinjian Wu, *Member, IEEE*

***Abstract*— Target detection becomes increasingly important in hyperspectral image (HSI) analysis but is limited by difficulties in acquiring accurate pixel-level training labels. This article proposes a multiple-instance metric learning neural network (MIML-Net) for hyperspectral target detection tasks, which only requires region-level labels and greatly alleviates the laborious pixel-level annotation problems. Our method learns the embeddings of regions with weak labels under attention-based multiple-instance learning (MIL) framework. Based on which, we impose a novel metric-based regularizer to constrain target and background embeddings to two learnable compact clusters with distinct centroids, which further boosts the spectral feature representation ability. The proposed metric-based regularizer enforces a discriminative detector due to its capability to reduce the intraclass variations and encourage the interclass separations simultaneously. Extensive experimental results from both simulated and real-field datasets demonstrate the effectiveness of the proposed MIML-Net in comparison with the state-of-the-art weakly supervised techniques.**

***Index Terms*— Discriminative feature embedding, hyperspectral, metric learning (ML), multiple-instance learning (MIL), target detection.**

I. Introduction

HYPERSPECTRAL images (HSIs) provide approximately contiguous spectral information over a wide range of the electromagnetic spectrum, which enables the detection and identification of land-cover targets by exploiting the subtle spectral discrepancies of materials. In the past decades, various applications of HSI have been developed and demonstrate promising performance, such as target detection [1], tree species classification [2], precision agriculture [3], and geological exploration [4].

Hyperspectral target detection is basically a binary classification task for spectra with the intention of distinguishing each pixel as a target or background [5]. The classical target detection approaches [6] are mainly based on spectral matching with prior knowledge of desired targets. Due to the spectral variability affected by the spectrometer noise and environmental conditions, such signature-based detectors are sensitive to the quality of the off-the-shelf target signatures in spectral library, resulting in unstable performance. It is also laborious to extract target signatures manually through visual inspection or global positioning systems (GPSs) for two reasons: 1) an analyst cannot visually discriminate subpixel targets and 2) there is always an offset between the GPS coordinate and the real location where the target appears. Thus, only regionwise labels can be assigned to indicate whether targets are included in an HSI region or not in practice [7], [8].

Multiple-instance learning (MIL), a framework of weakly supervised learning (WSL) [9], [10], was introduced to handle such ambiguously labeled data. Instead of receiving a set of pixelwise labeled instances in MIL, the learner receives datasets grouped as bags with bag-level labels, e.g., *positive* or *negative*. For a positive bag, it contains at least one target instance, while a negatively labeled bag is comprised entirely of nontargets. In practice, such binary labels can be conveniently assigned in two ways. First, if the GPS coordinates of the target are available, the areas surrounding the target can be considered as a positive bag, while the region beyond the GPS range of the target can be partitioned into negative bags. Second, target and nontarget areas can also be distinguished by the expert through visual examination or ground reference markers. The analyst can manually define bounding boxes that encompass regions with targets, treating these areas as positive bags without identifying the precise location of the target pixels. Regions without any apparent targets can be regarded as negative bags.

To address the target detection problems with such weak labels, many previous works focus on learning one or a group of discriminative target prototypes under the MIL framework, followed by a signature-based target detector. Extended functions of multiple instances (*e*FUMI) [11] aims to learn a target signature by maximizing the posterior statistics of weakly labeled bags using expectation–maximization (EM) approach. *e*FUMI performs well in recovering spectral characteristics, but it merges the bags with the same labels, which loses interbag and intrabag correlation. To alleviate the negative impacts of this problem, multiple-instance adaptive coherent detector (MI-ACE), multiple-instance spectral matched filter (MI-SMF) [12], and their extensions [13], [14] estimate one target concept that characterizes the target spectral features by maximizing the target detection response of each

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individual positive bag and minimizing the detection response on all negative bags. In order to tackle spectral variability problem, multiple-instance hybrid estimator (MI-HE) [15], multiple-instance multiple diverse adaptive coherent detector (MIMD-ACE) [16], and multitarget multiple-instance adaptive cosine estimator (MTMI-ACE) [17] are proposed to learn multiple target characterizations. Even though these aforementioned methods have achieved impressive performance, some inherent shortcomings still exist. Most of them need to follow handcrafted statistical assumptions, which are challenging to adapt to complex real-field scenarios. Besides, they rely heavily on shallow-level spectral features, which has been demonstrated to have limited representation capability.

Recently, the deep neural network (DNN) has shown explosive success in high-level discriminative feature learning [18] and has been applied in many hyperspectral target detection tasks. The MIL-based DNN are subsequently exploited but highly restricted by the unavoidable label ambiguity in the training data. Many scholars considered choosing a suitable permutation-invariant MIL pooling function [19], [20] to locate and aggregate the prime prototypes in weakly labeled bags, e.g., mean pooling, max pooling, and log-sum-exp pooling. The common shortcomings of those MIL pooling functions are that they are predefined and untrainable. Therefore, it would be desirable to consider a task-specific and adaptive MIL pooling function. Ilse et al. [21] proposed a deep attention-based aggregation operator served as a pooling function, giving the contribution of each instance to the bag label through trainable attention weights. Based on which, Jiao et al. [22] proposed an L_1 sparsity-regularized attention multiple-instance neural network (MINN), which imposes a sparsity constraint on attention weights to discriminate false-positive instances for hyperspectral target detection tasks. Those approaches have yielded significant performance but they pay relatively little attention to constraining or regularizing the embedding space to generate well-discriminated representations. They may suffer from handling intraclass variations, which is critical for training a robust detector. In particular, intraclass variations exist for three potential reasons: 1) since HSIs are susceptible to illumination conditions and sensor noise, the spectral characteristics of targets show great variabilities; 2) limited by the spatial resolution of the sensor, disparate materials might be mixed in one spectrum, resulting in subpixel targets and backgrounds; and 3) the background spectra are with natural diversities. It is essential to learn similar deep representations in feature space for targets with spectral variabilities and regularize background features to be compact. Moreover, the features of the targets and backgrounds should be separated as much as possible. In this situation, metric learning (ML) under the MIL framework is appealing.

To address the aforementioned difficulties and disadvantages, we propose a novel end-to-end multiple-instance ML neural network (MIML-Net) for hyperspectral target detection tasks, to selectively embed the relevant information from weakly labeled bags according to instance-level attention mechanism, and dynamically regularize the mappings of targets and nontargets to two learnable compact clusters with distinct centers. MIML-Net has three good properties: 1) learning bag embeddings using adaptive attention-based network can effectively extract *prime information* from HSI regions; 2) constraining the deep features of target (or background) into a compact cluster can facilitate the reduction of *intraclass variations*; and 3) increasing the distance between target and background features can enhance the *interclass separations*. MIML-Net jointly learns the well-discriminated embedding space and a pixelwise hyperspectral target detector with imprecise labels by formulating a hybrid loss, which consists of three well-designed parts: bag identification term, embedding similarity term, and centroid separation term. The learned target detector provides improved accuracy than previous contributions.

It is worth noting that, except for the proposed approach, ML under MIL framework for hyperspectral target detection tasks has been rarely investigated. The novelty and contributions of our work can be, therefore, summarized as follows.

1) To the best of our knowledge, multiple-instance ML (MIML) for hyperspectral target detection tasks is first introduced and explored in this article. Specifically, we utilize an MIL attention-based aggregation strategy to discriminate the key samples from hyperspectral regions and selectively fuse the spectral representations. Then, we explicitly model the metric relationships of MIL bag embeddings and regularize a discriminative feature space, which is conducive to spectral representation learning and brings better detection performance.
2) We propose a novel metric-based loss, encouraging target, and nontarget features to be close to two learnable separated centroids from the perspective of both Euclidean and angle distance. Leveraging the proposed metric-based loss, the learnable embeddings exhibit high intraclass compactness and interclass separability, thereby making the follow-up target detector more effective.
3) By utilizing the novel metric-based loss and the MIL attention aggregator, we build up MIML-Net for weakly supervised hyperspectral target detection. This preliminary work greatly eliminates the laborious pixel-level annotation problems and enhances the discrimination of spectral features simultaneously, which may inspire more research on this issue. Experiments on both simulated and real-field data compared with representative state-of-the-art approaches are provided to corroborate the effectiveness of the proposed structure.

The remainder of this article is organized as follows. In Section II, a brief review of related works under the MIL and ML framework is given, respectively. Section III demonstrates the details of the proposed MIML-Net. Section IV presents the experiments across both simulated and real-field datasets in comparison with classical and state-of-the-art weakly supervised techniques. Section V summarizes our contributions and outlines the future direction of our work in hyperspectral target detection field.

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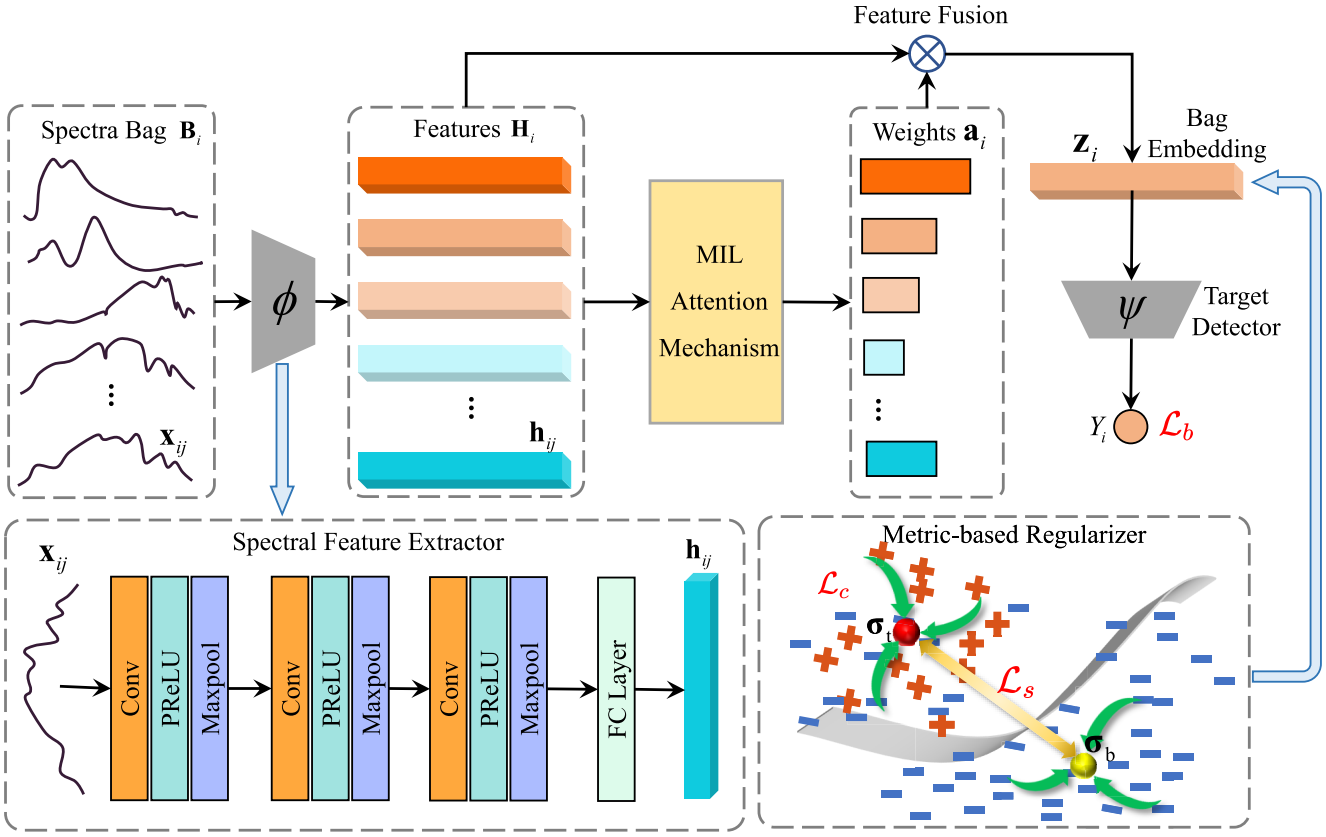

Fig. 1. Overall flowchart of the proposed MIML-Net. The spectra bags are feed into 1D-CNN-based feature extractor to get high-level spectral representation. Then, a learnable MIL attention module is utilized to discover and aggregate the prime information of bags. A novel metric-based regularizer is added to constrain the embedding space to generate well-discriminated representation, which makes the target detector more effective.

II. Related Work

A. Multiple-Instance Learning

MIL is first developed by Dietterich et al. [10] for drug activity prediction task and has achieved tremendous success in a variety of fields, such as pathology diagnosis [23], [24], object/anomaly detection [25], [26], [27], object tracking [28], image retrieval systems [29], and image captioning [30], in past few years. In MIL, the training data are partitioned into a set of bags with binary bag-level labels. In contrast to supervised machine learning methods, precise instance-level labels are not required in MIL framework, which greatly alleviates the difficulty arising from labeling objects.

Diverse density (DD) [9] and its improved version EM-DD [31] estimate a target concept by discovering the maximum intersection of positive bags and suppressing the union of negative bags. Multiple-instance support vector machine (mi-SVM) [32] iteratively trains an instance-level SVM classifier and update the hidden labels until the label assignment remains unchanged. Chen et al. [33] developed the MIL via embedded instance selection (MILES), which maps each bag into a discriminative feature space defined by selected instances and then build one-norm SVM for classification. MIL with randomized trees (MIForest) [34] defines the ground-truth labels in positive bags as random variables and iteratively training random forests through a fast iterative homotopy method for instance-level identification.

With the growth of deep learning, many scholars have attempted to use neural networks to tackle MIL problems. mi-Net and MI-Net [19] are two representative MIL-based deep learning approaches, which focus on the bag-level representation learning and prediction in an end-to-end learning fashion. Ilse et al. [21] first presented a learnable MIL pooling layer via neural networks, assigning the attention coefficients for each instance in weakly labeled bags to locate the hidden key information. In [35], a novel channel-spatial attention (CS-attention) module is utilized to mine the key features from both channel and spatial domain to better fuse the bag-level embeddings. Unlike the previous methods, Transformer-based correlated MIL (TransMIL) [23] follows non-independent and identical distribution (non-IID) hypothesis and is devised to explore correlation and contextual information between instances when making bag-level predictions in pathology whole slide image (WSI) diagnosis tasks. Double-tier feature distillation MIL (DTFD-MIL) [36] expands the number of bags by introducing the concept of pseudo-bags and builds a double-tier MIL technique for feature distillation, excelling on lung cancer WSI classification task.

B. Metric Learning

ML aims to establish a suitable similarity (or equivalently dissimilarity) measurement between training samples, which can enhance the capabilities of feature representation. The investigation of ML generally emphasizes on metric loss functions and has been proven to be effective in combination with DNN. Initially, the Siamese network along with a contrastive loss [37] was designed for face verification and has been

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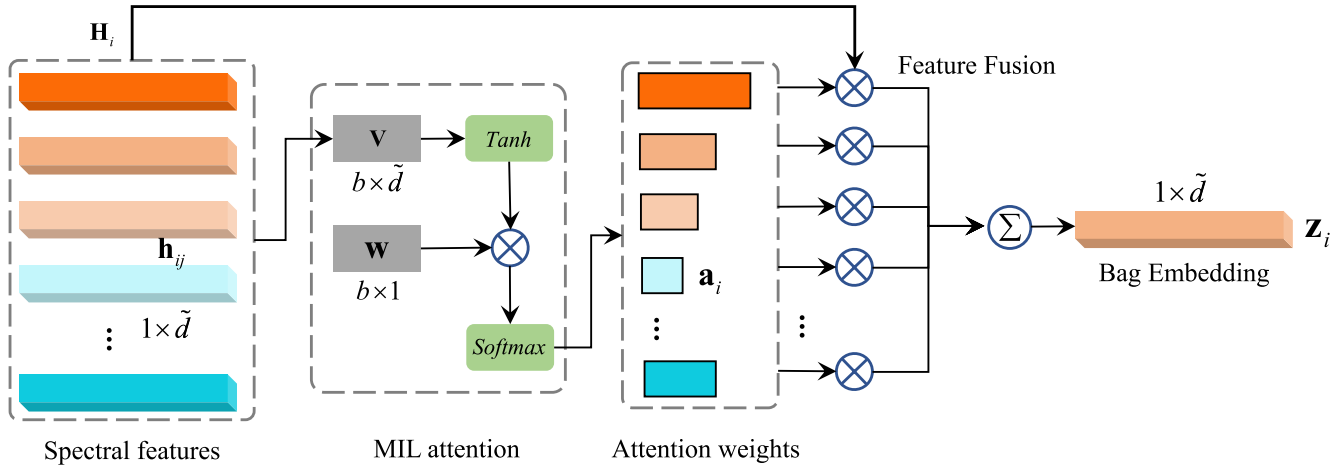

Fig. 2. MIL attention-based aggregation for discriminative feature learning. The embedding $\mathbf{z}_i$ of bag $\mathbf{B}_i$ is extracted by attention weight estimation and weighted feature fusion.

widely applied in many fields [38], [39]. This is accomplished by minimizing the following ranking loss objective:

$$\mathcal{L}_{\text{con}} = l\mathcal{D}^2(\mathbf{x}_i, \mathbf{x}_j)) + (1-l)(\max(0, \tau - \mathcal{D}(\mathbf{x}_i, \mathbf{x}_j)))^2 \quad (1)$$

where $\mathcal{D}(*, *)$ denotes the distance of pairwise input samples and τ is a margin value. $l \in \{0, 1\}$ is the label, which indicates whether the input data pairs $(\mathbf{x}_i, \mathbf{x}_j)$ are from the same class or not. The contrastive loss encourages that the sample pairs from the same class should be as close as possible; conversely, the sample pairs from different classes should be separated from each other. The lower bound of the distance is determined by τ. Inspired by the Siamese network, triplet network [40], [41] focuses on the similarity between pairwise samples from the same and different classes simultaneously, which yields

$$\mathcal{L}_{\text{tri}} = \max(0, \mathcal{D}(\mathbf{x}_i, \mathbf{x}_p) - \mathcal{D}(\mathbf{x}_i, \mathbf{x}_n) + \tau)^2 \quad (2)$$

where $(\mathbf{x}_i, \mathbf{x}_p)$ and $(\mathbf{x}_i, \mathbf{x}_n)$ denote the positive pairs and negative pairs. Specifically, $\mathbf{x}_p$ is from the same category as anchor instance $\mathbf{x}_i$, while $\mathbf{x}_n$ comes from a different category as $\mathbf{x}_i$. Quadruplet network [42] is modified based on a triplet network and has a better generalization ability and performance on the testing datasets. Angular loss [43] takes angle similarity into consideration between sample pairs. Those above-reviewed works can effectively capture the similarity of samples, but they have an obvious shortcoming that all of those approaches are needed to construct data pairs during the data preparation phase, which is tedious and costly.

Bearing this in mind, center loss [44] is proposed for dynamically bringing samples of the same class closer during the training stage without preconstructing sample pairs. It learns a center for deep representations of each class and penalizes the Euclidean distances between the features $\mathbf{x}_i$ and their corresponding class centers $\boldsymbol{c}_{y_i}$ as follows:

$$\mathcal{L}_{\text{center}} = \frac{1}{2}\sum_{i=1}^{m}\left\|\mathbf{x}_i - \boldsymbol{c}_{y_i}\right\|_2^2 \quad (3)$$

where m denotes the size of mini-batch. After that, constrained center loss [45] is proposed to enhance the intraclass compactness by introducing constraints on the magnitudes of the centers. Notably, even though center loss has been widely used to many popular tasks in recent years, such as action localization [46], person reidentification [47], anomaly detection [48], and 3-D object retrieval [49], center loss for hyperspectral target detection tasks remains almost undiscovered, especially for weakly supervised hyperspectral target detection. The previous contributions on center loss suffer from two main shortcomings, making them unsuitable for direct application to hyperspectral target detection. First, center loss requires fully supervised signals, while only imprecise labels in hyperspectral target detection are available in practice. Thus, a weakly supervised approach that can maintain the discriminative ability of ML is appealing. Second, center loss only enhances the compactness of intraclass representations. It does not consider the possible feature overlap between target and background features, which may greatly affect the discriminant ability of the detector.

III. Deep MIML-Net

In this section, we propose a deep MIML-Net to tackle the hyperspectral target detection problem with weak labels. The overall architecture of MIML-Net is illustrated in Fig. 1.

A. Problem Formulation

Let $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N] \in \mathbb{R}^{d\times N}$ be the training data, where d denotes the dimensionality of a spectrum and N is the total number of instances. The training data are grouped into K bags, $\mathbf{B} = \{\mathbf{B}_1, \ldots, \mathbf{B}_K\}$ with binary bag-level labels $L = [L_1, \ldots, L_K], L_i \in \{0, 1\}$. Specifically, there are K^+ positively labeled bags with associated annotation $L_i = 1, i = 1, \ldots, K^+$, and K^- negatively labeled bags with associated annotation $L_i = 0, i = K^+ + 1, \ldots, K$, where the total numbers of bags $K = K^+ + K^-$. A bag $\mathbf{B}_i$ contains N_i instances, which exhibit an unordered relationship. The total number of instances N is formulated as follows:

$$N = N^+ + N^- = \sum_{i=1}^{K^+} N_i + \sum_{i=K^++1}^{K} N_i \quad (4)$$

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where N^+ and N^- are the total number of positive instances and negative instances, respectively. Let $l_{ij} \in \{0, 1\}$ be the label of the jth instance in the ith bag. Notably, the ground-truth instance-level labels l_{ij} in positive bags $\mathbf{B}_i$ ($i = 1, \ldots, K^+$) are unknown. Since the MIL problem states that a positively labeled bag must contain at least one target instance and a negatively labeled bag is comprised entirely of nontargets, the relationship of bag-level labels L_i and instance-level labels l_{ij} can be indicated as follows:

$$L_i = \begin{cases} 0, & \text{iff } \sum l_{ij} = 0, j = 1, \ldots, N_i \\ 1, & \text{otherwise.} \end{cases} \tag{5}$$

B. Spectral Feature Extraction

Considering the impressive feature representation capacity of deep models, we utilize a 1-D convolutional neural network (1D-CNN) to extract the deep spectral features for each hyperspectral pixel. The architecture of 1D-CNN-based spectral feature extractor consists of three modules. Each module contains a convolutional layer with 1×3 kernel size, a parametric rectified linear unit (PReLU), and a max-pooling layer with 1×2 kernel size, respectively. Then, we map the features into fixed dimensional vectors using a fully connected (FC) layer. Define the spectral feature operator as $f(\mathbf{x}_{ij}; \boldsymbol{\phi})$, where $\boldsymbol{\phi}$ is the learnable parameter set. The spectral feature $\mathbf{h}_{ij}$ for the jth instance in ith bag is computed as follows:

$$\mathbf{h}_{ij} = f(\mathbf{x}_{ij}; \boldsymbol{\phi}) \tag{6}$$

where $\mathbf{h}_{ij}$ with dimension $1 \times \tilde{d}$. The feature set for each bag $\mathbf{B}_i$ is denoted as $\mathbf{H}_i = \{\mathbf{h}_1, \ldots, \mathbf{h}_{ij}, \ldots, \mathbf{h}_{iN_i}\} \in \mathbb{R}^{N_i \times \tilde{d}}$. $\tilde{d}$ is set to 500 in this configuration.

C. MIL Attention-Based Aggregation

In our proposed MIML-Net, the discriminative bag embeddings are extracted through iterative attention score estimation and weighted feature fusion. In practice, the attention weights are determined by a trainable neural network, which is parameterized by $\mathbf{w} \in \mathbb{R}^{b \times 1}$ and $\mathbf{V} \in \mathbb{R}^{b \times \tilde{d}}$. Denote the attention scores for each instance in $\mathbf{B}_i$ as $\mathbf{a}_i = [a_{i1}, \ldots, a_{ij}, \ldots, a_{iN_i}]$ with dimension $1 \times N_i$. Then, the attention weights predicted by the self-attention network are indicated as follows:

$$a_{ij} = \frac{\exp\left\{\mathbf{w}^{\mathrm{T}} \tanh\left(\mathbf{V}\mathbf{h}_{ij}^{\mathrm{T}}\right)\right\}}{\sum_{j=1}^{N_i} \exp\left\{\mathbf{w}^{\mathrm{T}} \tanh\left(\mathbf{V}\mathbf{h}_{ij}^{\mathrm{T}}\right)\right\}}. \tag{7}$$

The attention coefficients for a specific bag are subject to sum-to-one and nonnegative constraints, which can be denoted as follows:

$$\begin{aligned} &\text{sum-to-one: } \mathbf{a}_i^{\mathrm{T}}\mathbf{1} = 1 \\ &\text{nonnegativity: } \mathbf{a}_i \succeq 0 \end{aligned} \tag{8}$$

where $\mathbf{1}$ is a vector with all elements 1 and $\succeq$ denotes componentwise inequality. These two constraints can be guaranteed by a softmax layer. Then, the fused bag embedding $\mathbf{z}_i$ of the ith bag is computed as the weighted average of the instance-level deep features $\mathbf{H}_i$ and attentions scores $\mathbf{a}_i$, which yields

$$\mathbf{z}_i = \sum_{j=1}^{N_i} a_{ij}\mathbf{h}_{ij}. \tag{9}$$

Fig. 2 shows the corresponding flowchart of MIL attention-based aggregation. MIL attention-based aggregation eliminates the uncertainty from region-level labels. It plays a critical role in discriminating the salient target samples from positive bags and aggregating the prime background information from negative bags. After aggregation operation, the bag embeddings of positive and negative bags could explicitly represent target and nontarget spectral information, respectively.

D. Discriminative MIML

Discriminative MIML contains three main components, known as bag identification term $\mathcal{L}_b$, embedding similarity term $\mathcal{L}_c$, and centroid separation term $\mathcal{L}_s$.

1) Bag Identification: After bag-level feature aggregation, we build up a binary classifier upon fusion representations for prediction. In practice, the fused embeddings are fed into an FC layer $\mathcal{F}(\mathbf{z}_i; \boldsymbol{\psi}, b)$ with single output along with a sigmoid function $\text{Sigmoid}(*) = 1/(1 - \exp(*))$ to indicate the probability of an embedding to be positive or negative, shown as follows:

$$Y_i = \mathcal{F}(\mathbf{z}_i; \boldsymbol{\psi}, b) \tag{10}$$

$$= \frac{1}{1 - \exp\left(\boldsymbol{\psi}^{\mathrm{T}}\mathbf{z}_i + b\right)} \tag{11}$$

$$= \frac{1}{1 - \exp\left(\boldsymbol{\psi}^{\mathrm{T}}\left(\sum_{j=1}^{N_i} a_{ij} f(\mathbf{x}_{ij}, \boldsymbol{\phi})\right) + b\right)} \tag{12}$$

where $\boldsymbol{\psi}$ and b are the learnable weights and bias of the FC layer, respectively,[1] and Y_i is the probability prediction. Considering bag-level label L in (5) is a binary random variable [21], Bernoulli distribution is applied to model the probability of L given a bag of instance $\mathbf{B}$, as follows:

$$p(L|\mathbf{B}) = \mathcal{F}(\mathbf{z}; \boldsymbol{\psi})^L (1 - \mathcal{F}(\mathbf{z}; \boldsymbol{\psi}))^{(1-L)}. \tag{13}$$

Then, we maximize the joint probability of all training bags by minimizing the sum of negative logarithm of Bernoulli distribution in (13), shown as follows:

$$\mathcal{L}_b = -\sum_{i=1}^{K^+} L_i \log(\mathcal{F}(\mathbf{z}_i; \boldsymbol{\psi})) - \sum_{i=K^+ +1}^{K} (1 - L_i) \log(1 - \mathcal{F}(\mathbf{z}_i; \boldsymbol{\psi})) \tag{14}$$

where $\mathcal{L}_b$ represents the bag identification loss.

2) Embedding Similarity: HSIs are easily affected by the imaging environment and equipment, yielding target spectra with high variability. Also, the background spectra show great diversities, because imaging scenes are often covered by various materials. Capturing intraclass spectral variations is a crucial factor to accurately detect the target pixels and subpixels hidden in complex background scenes. For this purpose, it is natural to consider the similarities of bag-level embeddings to reduce the variations of the target (or background) in the feature space. Inspired by the previous contribution *center loss* [44], we intend to cluster bag embeddings into

[1]For the sake of brevity, the bias b of the detector has been omitted in the following statement and equations, i.e., $\mathcal{F}(\mathbf{z}_i; \boldsymbol{\psi}, b) \longrightarrow \mathcal{F}(\mathbf{z}_i; \boldsymbol{\psi})$.

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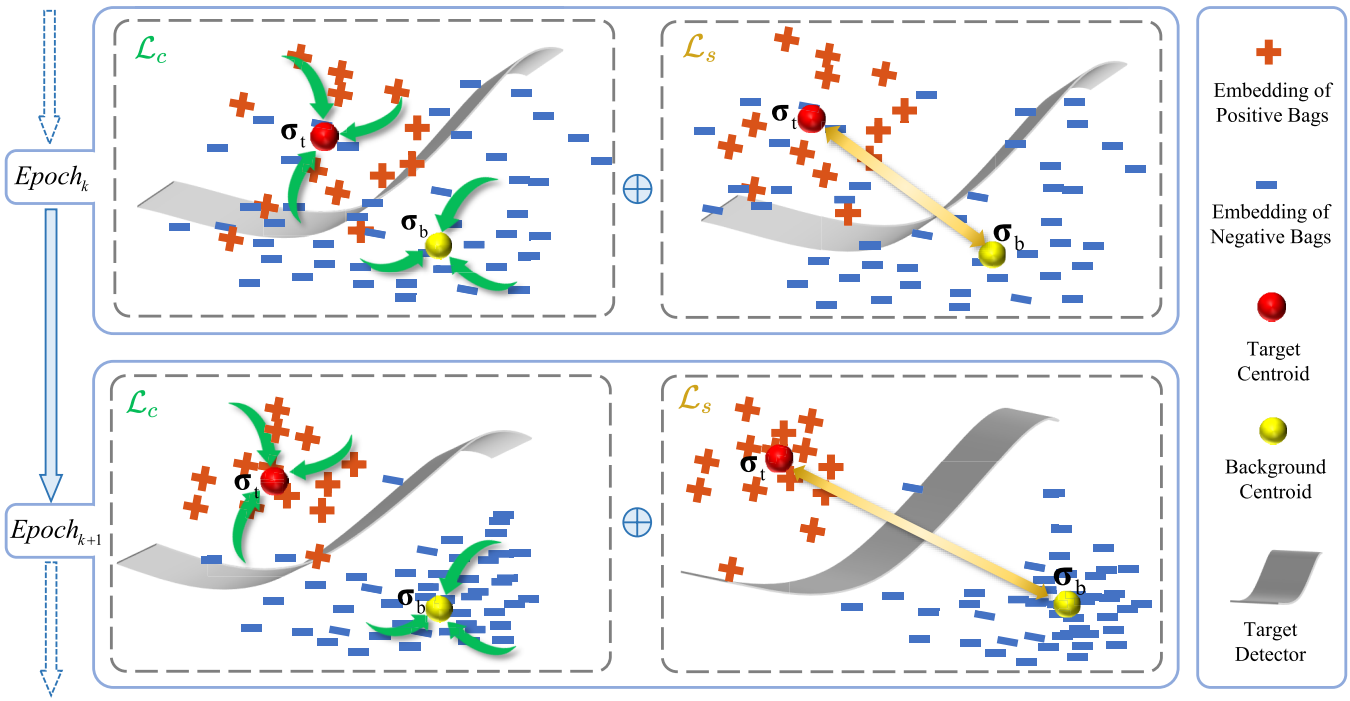

Fig. 3. Illustration of the proposed metric-based regularizer. In each iteration, the embedding similarity term $\mathcal{L}_c$ is employed to constrain the bag embedding of target or background to be as compact as possible. Also, the centroid separation term $\mathcal{L}_s$ is utilized to facilitate the separation of target and background. With end-to-end optimization, the embeddings enjoy high intraclass compactness and interclass separability.

two separate learnable centroids, e.g., target centroid $\boldsymbol{\sigma}_t$ and background centroid $\boldsymbol{\sigma}_b$, to regularize the fusion representations, shown as follows:

$$\mathcal{L}_c = \frac{1}{2}\sum_{i=1}^{K^+}\|\mathbf{z}_i - \boldsymbol{\sigma}_t\|_2^2 + \frac{1}{2}\sum_{i=K^++1}^{K}\|\mathbf{z}_i - \boldsymbol{\sigma}_b\|_2^2. \tag{15}$$

In (15), $\boldsymbol{\sigma}_t$ and $\boldsymbol{\sigma}_b$ are initialized from a standard normal distribution (SND) with the same dimension as bag embeddings $\mathbf{z}_i$, i.e., $\boldsymbol{\sigma}_{ti} \sim \mathcal{N}(0,1)$ and $\boldsymbol{\sigma}_{bi} \sim \mathcal{N}(0,1)$, and jointly optimized with the aforementioned $\mathcal{L}_b$ and the follow-up $\mathcal{L}_s$ in an end-to-end manner. By minimizing this term, MIML-Net tends to extract similar target embeddings from different positive bags while penalizing background diversity in the feature space.

3) Centroid Separation: The embedding similarity loss $\mathcal{L}_c$ reduces the variations of the target (or background) representations $\mathbf{z}_i$, which alleviates the intraclass spectral inconsistencies. However, it is inadequate for solely enhancing the compactness of target and background features. Due to the low spatial resolution of HSIs, there are lots of mixed pixels, inevitably resulting in a potential overlap between the spectral distributions of target and background [50], [51]. Neither embedding similarity modeling nor center loss [see (3)] takes into account the possible feature overlap. Thus, the separability between the target and background pixels in deep feature space should be investigated. We further utilize a centroid separation term, which is identified by the cosine similarity between two learnable centroids to supplement the embedding similarity term, formulated as follows:

$$\mathcal{L}_s = \frac{1}{\pi}\arccos\frac{\boldsymbol{\sigma}_t^{\mathrm{T}}\boldsymbol{\sigma}_b}{\|\boldsymbol{\sigma}_t\|_2\|\boldsymbol{\sigma}_b\|_2} \tag{16}$$

$$= \frac{1}{\pi}\arccos\hat{\boldsymbol{\sigma}}_t^{\mathrm{T}}\hat{\boldsymbol{\sigma}}_b \tag{17}$$

where $\hat{\boldsymbol{\sigma}}_t = \boldsymbol{\sigma}_t/\|\boldsymbol{\sigma}_t\|_2$, $\hat{\boldsymbol{\sigma}}_b = \boldsymbol{\sigma}_b/\|\boldsymbol{\sigma}_b\|_2$, and $\arccos(*)$ is the inverse cosine function with a range of $[0, \pi]$. Intuitively, the target cluster and the background cluster should be separated from each other. Thus, we explicitly impose the center embedding angle as the separation regularization, which enforces the representations of target and background to be mutually distant from each other. Moreover, the constant $1/\pi$ is designed to normalize the range of angle information (arccos) to $[0, 1]$. Centroid separation loss $\mathcal{L}_s$ enables the features of target more distinctive in contrast to backgrounds. Compared with center loss, the proposed metric-based loss can not only reduce the intraclass variations, but also increase interclass separations. It explicitly regularizes a discriminative feature space from the view of both Euclidean distance and angle information. The interpretation of the proposed metric-based regularizer can also be visually illustrated in Fig. 3.

E. Overall Architecture and Optimization

The overall loss function of MIML-Net is given as follows:

$$\mathcal{L} = \mathcal{L}_b + \underbrace{\lambda\mathcal{L}_c + \eta\mathcal{L}_s}_{\text{metric-based loss}} \tag{18}$$

where λ and η are two hyperparameters to balance three terms. Our network is optimized by the standard Adam optimizer [52]. Unlikely metric-based Siamese [37] or triplet [40]

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network, our formulation circumvents the drawbacks of formulating the data pairs, which is more efficient.

The learnable parameters of MIML-Net can be divided into three parts, $\mathcal{W} = \{\boldsymbol{\phi}, \mathbf{w}, \mathbf{V}\}$ for attention-based feature extraction and fusion, $\boldsymbol{\psi}$ for target detector, and two centroids $\boldsymbol{\sigma} = \{\boldsymbol{\sigma}_t, \boldsymbol{\sigma}_b\}$ for ML. The gradient of the overall loss function with respect to one bag embedding $\mathbf{z}_i$ at kth iteration can be calculated by

$$\frac{\mathrm{d}\mathcal{L}^k}{\mathrm{d}\mathbf{z}_i^k} = \frac{\mathrm{d}\mathcal{L}_b^k}{\mathrm{d}\mathbf{z}_i^k} + \lambda\frac{\mathrm{d}\mathcal{L}_c^k}{\mathrm{d}\mathbf{z}_i^k} + \eta\frac{\mathrm{d}\mathcal{L}_s^k}{\mathrm{d}\mathbf{z}_i^k}. \tag{19}$$

Specifically, the partial derivative of embedding similarity term $\mathcal{L}_c$ with respect to $\mathbf{z}_i$ can be computed as follows:

$$\frac{\mathrm{d}\mathcal{L}_c}{\mathrm{d}\mathbf{z}_i} = \delta(L_i = 1)(\mathbf{z}_i - \boldsymbol{\sigma}_t) + \delta(L_i = 0)(\mathbf{z}_i - \boldsymbol{\sigma}_b) \tag{20}$$

where the Kronecker delta function $\delta(\mathcal{C}) = 1$ if the condition $\mathcal{C}$ is met; otherwise, $\delta(\mathcal{C}) = 0$. The partial derivative of centroid separation term $\mathcal{L}_s$ with respect to $\mathbf{z}_i$ is $\mathrm{d}\mathcal{L}_s/\mathrm{d}\mathbf{z}_i = 0$.[2]

Two centroids $\boldsymbol{\sigma}_t$ and $\boldsymbol{\sigma}_b$ can be updated iteratively in each mini-batch. The partial derivative of embedding similarity term $\mathcal{L}_c$ with respect to $\boldsymbol{\sigma} = \{\boldsymbol{\sigma}_t, \boldsymbol{\sigma}_b\}$ is given as follows:

$$\frac{\mathrm{d}\mathcal{L}_c}{\mathrm{d}\boldsymbol{\sigma}} = \delta(L_i = 1)(\boldsymbol{\sigma}_t - \mathbf{z}_i) + \delta(L_i = 0)(\boldsymbol{\sigma}_b - \mathbf{z}_i). \tag{21}$$

Algorithm 1 MIML-Net Algorithm

Input: MIL training data bags $\mathbf{B} = \{\mathbf{B}_1, \ldots, \mathbf{B}_K\}$ with corresponding labels $L = [L_1, \ldots, L_K]$. Batch size m. Learning rate μ_1, μ_2 for Adam optimizer. Loss balance factor λ and η. Maximum training epoch M.

Output: $\mathcal{W} = \{\boldsymbol{\phi}, \mathbf{w}, \mathbf{V}\}$, $\boldsymbol{\psi}$, and $\boldsymbol{\sigma} = \{\boldsymbol{\sigma}_t, \boldsymbol{\sigma}_b\}$.

Initialize parameters $\mathcal{W} = \{\boldsymbol{\phi}, \mathbf{w}, \mathbf{V}\}$ for attention-based feature extraction and fusion, $\boldsymbol{\psi}$ for target detector, and two centroids $\boldsymbol{\sigma} = \{\boldsymbol{\sigma}_t, \boldsymbol{\sigma}_b\}$ for metric learning. Current iteration number $k = 0$.
repeat
  Extract instance-level spectral representation, $\mathbf{h}_{ij}$, in training bags, according to Eq. (6)
  Calculate the attention weights, a_{ij}, for each instance, according to Eq. (7)
  Extract the fusion feature, $\mathbf{z}_i$, for each training bag, according to Eq. (9)
  Compute the total loss, $\mathcal{L}$, according to Eq. (14)-(18)
  Compute the backpropagation gradient, $\mathrm{d}\mathcal{L}^k/\mathrm{d}\mathbf{z}_i^k$, for $\mathbf{z}_i$, according to Eq. (19)
  Update $\boldsymbol{\psi}$, according to Eq. (27)
  Update centroids $\boldsymbol{\sigma} = \{\boldsymbol{\sigma}_t, \boldsymbol{\sigma}_b\}$, according to Eq. (31) and Eq. (32), respectively
  Update $\mathcal{W} = \{\boldsymbol{\phi}, \mathbf{w}, \mathbf{V}\}$, according to Eq. (28)
  $k \leftarrow k + 1$
until Maximum training epoch reached

[2]The partial derivative of bag identification term $\mathcal{L}_b$ with respect to $\mathbf{z}_i$, $(\mathrm{d}\mathcal{L}_b/\mathrm{d}\mathbf{z}_i)$, can be treated as a straightforward derivative of the standard binary cross-entropy (BCE) loss. We omit it for simplicity here.

As for centroid separation term $\mathcal{L}_s$, the gradient of $\mathcal{L}_s$ with respect to $\boldsymbol{\sigma}_t$ is shown as follows:

$$\frac{\mathrm{d}\mathcal{L}_s}{\mathrm{d}\boldsymbol{\sigma}_t} = \frac{1}{\pi}\frac{\mathrm{d}\left(\arccos\hat{\boldsymbol{\sigma}}_t^{\mathrm{T}}\hat{\boldsymbol{\sigma}}_b\right)}{\mathrm{d}\left(\hat{\boldsymbol{\sigma}}_t^{\mathrm{T}}\hat{\boldsymbol{\sigma}}_b\right)}\frac{\mathrm{d}\left(\hat{\boldsymbol{\sigma}}_t^{\mathrm{T}}\hat{\boldsymbol{\sigma}}_b\right)}{\mathrm{d}\boldsymbol{\sigma}_t} \tag{22}$$

$$= -\frac{1}{\pi\sqrt{1-\left(\hat{\boldsymbol{\sigma}}_t^{\mathrm{T}}\hat{\boldsymbol{\sigma}}_b\right)^2}}\left(\frac{\mathbf{I}}{\|\boldsymbol{\sigma}_t\|_2} - \frac{\boldsymbol{\sigma}_t\boldsymbol{\sigma}_t^{\mathrm{T}}}{\|\boldsymbol{\sigma}_t\|_2^3}\right)\hat{\boldsymbol{\sigma}}_b \tag{23}$$

$$= -\frac{1}{\pi\sqrt{1-\left(\hat{\boldsymbol{\sigma}}_t^{\mathrm{T}}\hat{\boldsymbol{\sigma}}_b\right)^2}}\frac{\mathbf{I} - \hat{\boldsymbol{\sigma}}_t\hat{\boldsymbol{\sigma}}_t^{\mathrm{T}}}{\|\boldsymbol{\sigma}_t\|_2}\hat{\boldsymbol{\sigma}}_b \tag{24}$$

$$= -\frac{\left(\mathbf{I} - \hat{\boldsymbol{\sigma}}_t\hat{\boldsymbol{\sigma}}_t^{\mathrm{T}}\right)\hat{\boldsymbol{\sigma}}_b}{\pi\|\boldsymbol{\sigma}_t\|_2\sqrt{1-\left(\hat{\boldsymbol{\sigma}}_t^{\mathrm{T}}\hat{\boldsymbol{\sigma}}_b\right)^2}} \tag{25}$$

where $\mathbf{I}$ is the identity matrix. Similarly, $\mathrm{d}\mathcal{L}_s/\mathrm{d}\boldsymbol{\sigma}_b$ can be written as follows:

$$\frac{\mathrm{d}\mathcal{L}_s}{\mathrm{d}\boldsymbol{\sigma}_b} = -\frac{\left(\mathbf{I} - \hat{\boldsymbol{\sigma}}_b\hat{\boldsymbol{\sigma}}_b^{\mathrm{T}}\right)\hat{\boldsymbol{\sigma}}_t}{\pi\|\boldsymbol{\sigma}_b\|_2\sqrt{1-\left(\hat{\boldsymbol{\sigma}}_t^{\mathrm{T}}\hat{\boldsymbol{\sigma}}_b\right)^2}}. \tag{26}$$

Then, we have the following updating equations in each mini-batch with batch size m:

$$\boldsymbol{\psi}^{k+1} = \boldsymbol{\psi}^{k+1} - \mu_1\frac{\mathrm{d}\mathcal{L}_b^k}{\mathrm{d}\boldsymbol{\psi}^k} \tag{27}$$

$$\mathcal{W}^{k+1} = \mathcal{W}^k - \mu_1\frac{\mathrm{d}\mathcal{L}^k}{\mathrm{d}\mathbf{z}_i^k}\frac{\mathrm{d}\mathbf{z}_i^k}{\mathrm{d}\mathcal{W}^k} \tag{28}$$

$$\boldsymbol{\sigma}_t^{k+1} = \boldsymbol{\sigma}_t^k - \mu_2\Delta\boldsymbol{\sigma}_t^k \tag{29}$$

$$\boldsymbol{\sigma}_b^{k+1} = \boldsymbol{\sigma}_b^k - \mu_2\Delta\boldsymbol{\sigma}_b^k \tag{30}$$

where μ_1 is the learning rate for $\mathcal{W}$ and Ψ, and μ_2 is the learning rate for centroids $\boldsymbol{\sigma}$. The gradients of centroids $\Delta\boldsymbol{\sigma}_t$ and $\Delta\boldsymbol{\sigma}_t$ are computed as follows:

$$\Delta\boldsymbol{\sigma}_t = \frac{\sum_{i=1}^m \delta(L_i = 1)(\boldsymbol{\sigma}_t - \mathbf{z}_i)}{\epsilon + \sum_{i=1}^m \delta(L_i = 1)} - \frac{\eta}{\lambda}\cdot\frac{\left(\mathbf{I} - \hat{\boldsymbol{\sigma}}_t\hat{\boldsymbol{\sigma}}_t^{\mathrm{T}}\right)\hat{\boldsymbol{\sigma}}_b}{\pi\|\boldsymbol{\sigma}_t\|_2\sqrt{1-\left(\hat{\boldsymbol{\sigma}}_t^{\mathrm{T}}\hat{\boldsymbol{\sigma}}_b\right)^2}} \tag{31}$$

$$\Delta\boldsymbol{\sigma}_b = \frac{\sum_{i=1}^m \delta(L_i = 0)(\boldsymbol{\sigma}_b - \mathbf{z}_i)}{\epsilon + \sum_{i=1}^m \delta(L_i = 0)} - \frac{\eta}{\lambda}\cdot\frac{\left(\mathbf{I} - \hat{\boldsymbol{\sigma}}_b\hat{\boldsymbol{\sigma}}_b^{\mathrm{T}}\right)\hat{\boldsymbol{\sigma}}_t}{\pi\|\boldsymbol{\sigma}_b\|_2\sqrt{1-\left(\hat{\boldsymbol{\sigma}}_t^{\mathrm{T}}\hat{\boldsymbol{\sigma}}_b\right)^2}} \tag{32}$$

where ϵ is a small positive tuning number in case the denominator is zero. The MIML-Net can be summarized as in Algorithm 1. During the testing process, a spectrum is first fed into the 1D-CNN-based feature extractor to obtain the deep spectral representation. Then, we skip the MIL attention-based aggregation module, because the attention weight defaults to 1.0 for a single-pixel input. The spectral features are directly passed through the well-trained target detector to get the detection response.

IV. Experiments

In this section, we evaluate our method MIML-Net in comparison with classical and state-of-the-art weakly supervised techniques on both simulated and real-field hyperspectral data.

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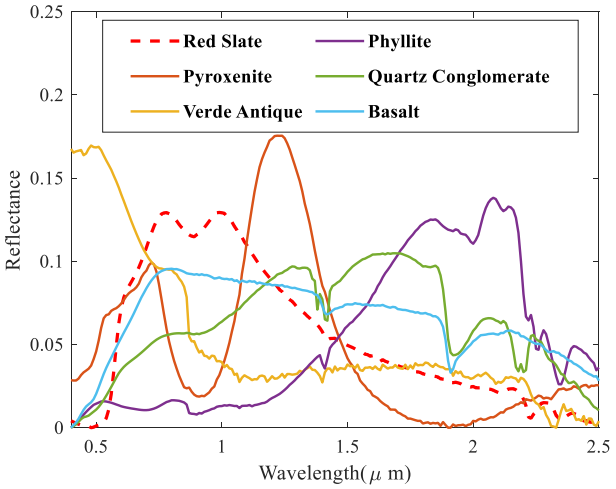

Fig. 4. Selected end-members from ASTER spectral library. Specifically, Red Slate spectrum (indicated by red dashed line) was assigned as the target end-member, and Basalt, Pyroxenite, Verde Antiquez, Phyllite, and Quartz Conglomerate were specified as background end-members.

TABLE I

AREA UNDER THE ROC CURVES (AUCS) FOR MIML-NET AND COMPARISON ALGORITHMS ON SIMULATED HYPERSPECTRAL DATA WITH DIFFERENT MEAN TARGET PROPORTIONS. AVERAGE AUCS ARE SHOWN IN THE LAST COLUMN. BEST RESULTS ARE SHOWN IN BOLD, AND SECOND BEST RESULTS ARE UNDERLINED

Alg.	$\bar{p}_t$			Avg. AUC
	0.1	0.2	0.3	
EM-DD [31]	0.658	0.742	0.834	0.745
MI-ACE [12]	0.671	0.752	0.839	0.754
MI-SMF [12]	0.672	0.753	0.840	0.755
MILES [33]	0.744	0.775	0.786	0.768
MI-HE [15]	0.707	0.778	0.832	0.772
MIMD-ACE [16]	0.668	0.751	0.837	0.752
MIMD-SMF [16]	0.660	0.737	0.829	0.742
Max-attention MINN [53]	0.751	0.799	0.812	0.787
CS-attention MINN [35]	0.758	0.835	0.847	0.813
L1-attention MINN [22]	0.739	0.827	0.849	0.805
Attention-DMIL [21]	0.737	0.824	0.836	0.799
MIML-Net (proposed)	**0.786**	**0.857**	**0.882**	**0.841**

Analysis and discussion based on experimental performance are given to further illustrate the effectiveness of MIML-Net.

A. Experimental Setup

The comparison algorithms deployed in this article include EM-DD [31], MILES [33], MI-ACE [12], MI-SMF [12], MI-HE [15], MIMD-ACE [16], MIMD-SMF [16], max-attention MINN [53], attention-DMIL [21], CS-attention MINN [35], and $L1$-attention MINN [22]. We implemented our method with Intel i7-10700k CPU and a single NVIDIA GeForce RTX 3080 GPU, under the PyTorch environment. We empirically set the hyperparameters batch size $m = 128$ for simulated data and $m = 64$ for Gulfport data, learning rate $\mu_1 = 1 \times 10^{-3}$, $\mu_2 = 0.5$, loss balance factor $\lambda = 2 \times 10^{-3}$ for simulated data and $\lambda = 5 \times 10^{-3}$ for Gulfport data, $\eta = 1 \times 10^{-2}$, and maximum training epoch $M = 600$. The receiver operating characteristic curves (ROCs) specified by false positive rate (FPR) and true positive rate (TPR), and the area under the curve (AUC) were used for performance evaluation metrics.

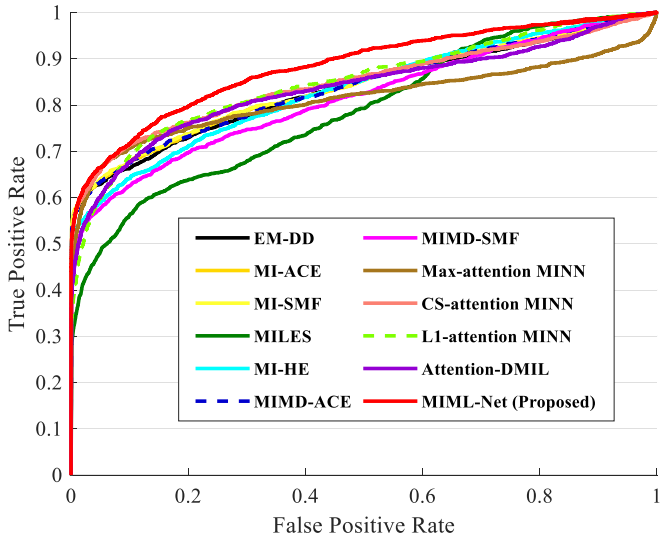

Fig. 5. ROC curves on training data with mean target proportions $\bar{p}_t = 0.3$.

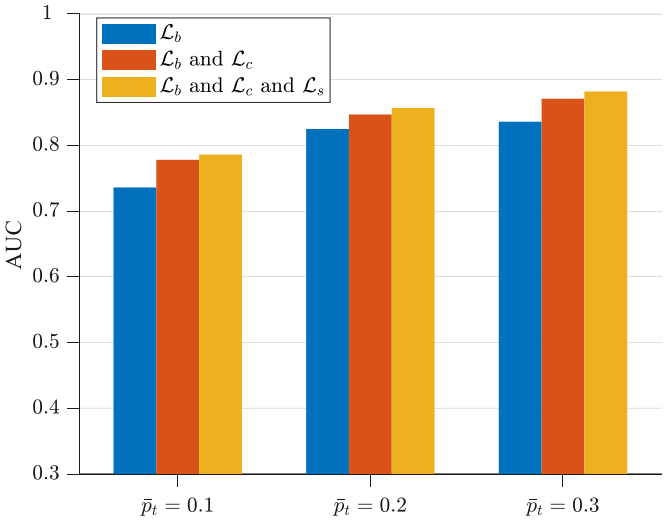

Fig. 6. Ablation study over three simulated dataset with varied mean target proportions $\bar{p}_t$ using different combinations of three loss terms.

B. Simulated Dataset

We picked six spectra from the advanced spaceborne thermal emission reflection radiometer (ASTER) spectral library [54] as shown in Fig. 4 to produce the highly mixed simulated data. Specifically, Red Slate spectrum was assigned as the target end-member, and Basalt, Pyroxenite, Verde Antiquez, Phyllite, and Quartz Conglomerate were specified as background end-members. The proportions of end-members are sampled from the Dirichlet distribution and mixed following the linear mixing model. Gaussian white noise is added to control the signal-to-noise ratio (SNR) of synthetic signatures to 30 dB. Three highly mixed sets of simulated data with different average target proportions were yielded. For each set, there are 300 positive bags and 700 negative bags, respectively. It contains 25 instances for each bag. There are five instances containing target components for each positive bag. The mean target proportions $\bar{p}_t$ are set to 0.1, 0.2, and 0.3 for three sets of simulated data, respectively. The simulated procedure and

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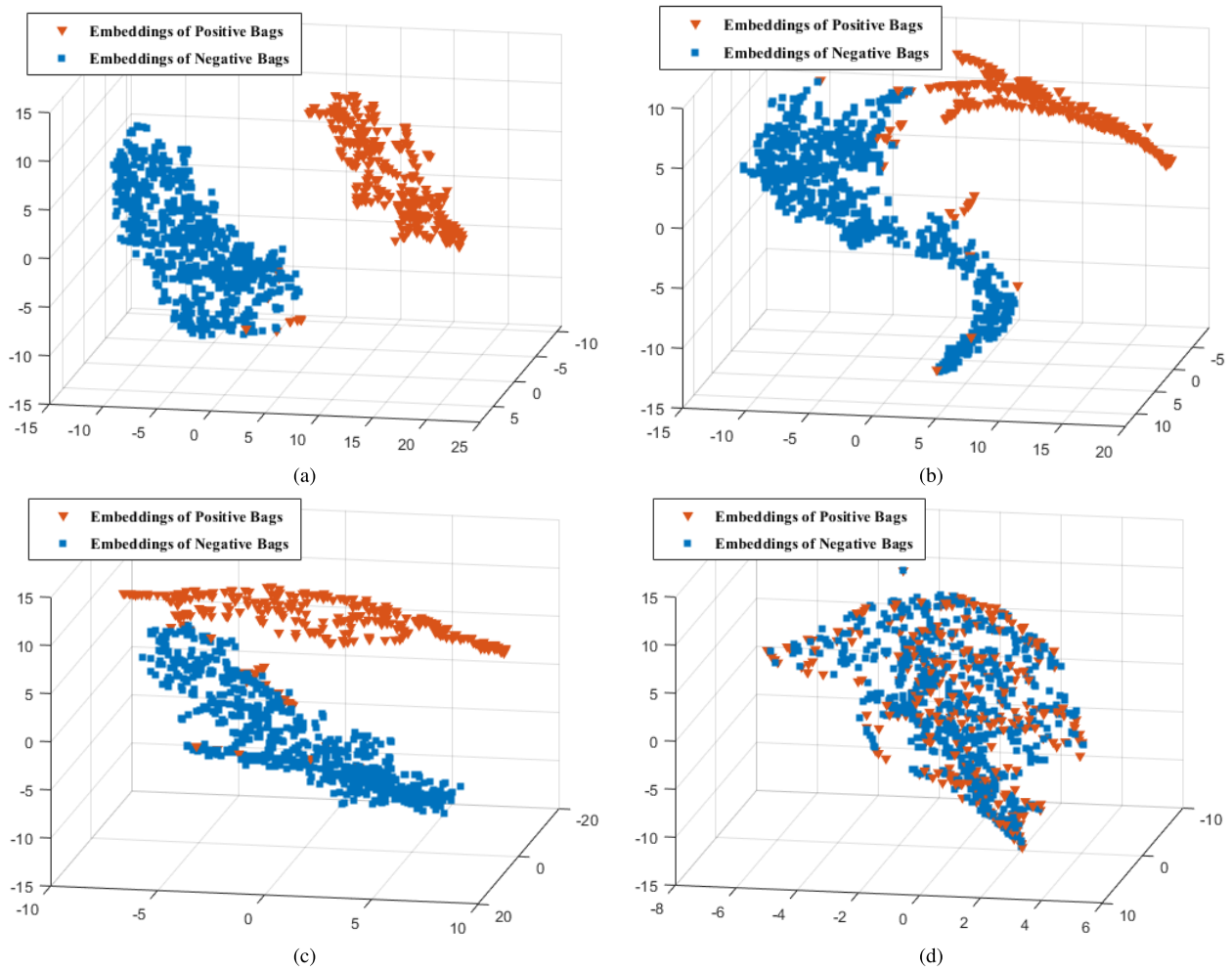

Fig. 7. 3-D t-SNE visualization of embeddings on simulated dataset with $\bar{p}_t = 0.3$ for the proposed MIML-Net and comparison methods. (a) MIML-Net. (b) Attention-DMIL. (c) $L1$-attention MINN. (d) Max-attention MINN. Orange triangles and blue squares denote the learned embeddings of positive and negative bags, respectively.

code[3] are available in [11]. Test data and validation data are produced in the same fashion.

1) Performance Evaluation: We evaluated MIML-Net with comparison techniques on aforementioned simulated datasets. The experimental results are reported in Table I. From the perspective of AUC values, MIML-Net significantly outperforms the comparison methods on all three highly mixed subpixel target detection tasks. The ROC curves on training data with mean target proportions $\bar{p}_t = 0.3$ are shown in Fig. 5. We can still find that the ROC curve of MIML-Net is evidently higher than others, which means better overall performance.

2) Ablation Study: In order to verify the importance of proposed metric-based regularizer, we conduct an ablation study over three simulated datasets with varied mean target proportions $\bar{p}_t$ using different combinations of three loss terms. We compared the experimental results in three cases, respectively: 1) using only $\mathcal{L}_b$; 2) using $\mathcal{L}_b$ and $\mathcal{L}_c$; and 3) using $\mathcal{L}_b$, $\mathcal{L}_c$, and $\mathcal{L}_s$ simultaneously, as shown in Fig. 6 and Table II. It can be clearly observed that the performance gradually improves from case 1 to case 3 on all three datasets, which provides solid evidence to effectiveness of the regularizer in our methods. We argue that MIML-Net leverages the proposed metric-based regularizer to confine the target and background features to two learnable compact clusters with independent centroids, which is beneficial to detection tasks.

The running times of the three cases are recorded in the ablation study in Table II. The rightmost column of Table II lists the average forward inference time t per mini-batch with size $m = 128$ and the increased time cost Δt of the current case compared with the previous case. It could be noticed that the proposed metric-based loss ($\mathcal{L}_c$ and $\mathcal{L}_s$) only consumes a small amount of extra inference time, mainly owing to the two additional learnable centroids $\boldsymbol{\sigma}_t$ and $\boldsymbol{\sigma}_b$.

3) 3-D t-SNE Visualization of Embeddings: To show the effectiveness of feature regularization, we explicitly display the bag embeddings of training dataset. Fig. 7 shows the 3-D t-distributed stochastic neighbor embedding (t-SNE) [55] visualization of bag embeddings on simulated dataset with $\bar{p}_t = 0.3$ for MIML-Net and several comparison methods. Orange triangles and blue squares denote the learned embeddings of positive and negative bags, respectively. We can observe that the proposed MIML-Net learns more compact

[3]https://github.com/GatorSense/Hyperspectral_Data_Simulation

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TABLE II

ABLATION STUDY OVER THREE SIMULATED DATASET WITH VARIED MEAN TARGET PROPORTIONS $\bar{p}_t$ USING DIFFERENT COMBINATIONS OF THREE LOSS TERMS. BEST RESULTS ARE SHOWN IN BOLD

Case Index	$\mathcal{L}_b$	$\mathcal{L}_c$	$\mathcal{L}_s$	$\bar{p}_t$			Avg. AUC	Running Time/Batch $t(+\Delta t)$
				0.1	0.2	0.3		
1	✓	×	×	0.737	0.824	0.836	0.799	0.0079s
2	✓	✓	×	0.778	0.847	0.871	0.832	0.0088s (+0.0009s)
3	✓	✓	✓	**0.786**	**0.857**	**0.882**	**0.841**	0.0091s (+0.0003s)

TABLE III

AREA UNDER THE ROC CURVES (AUCS) OF MIML-NET USING DIFFERENT CENTROID INITIALIZATION METHODS ON SIMULATED HYPERSPECTRAL DATA WITH DIFFERENT MEAN TARGET PROPORTIONS. AVERAGE AUCS ARE SHOWN IN THE LAST COLUMN. BEST RESULTS ARE SHOWN IN BOLD, AND SECOND BEST RESULTS ARE UNDERLINED

Centroid Init. Methods	$\bar{p}_t$			Avg. AUC
	0.1	0.2	0.3	
SND Init.	**0.786**	<u>0.857</u>	<u>0.882</u>	**0.841**
Xavier Init. [56]	0.771	**0.859**	0.861	0.830
He Init. [57]	<u>0.783</u>	0.846	**0.885**	<u>0.838</u>

TABLE IV

AREA UNDER THE ROC CURVES (AUCS) OF MIML-NET USING DIFFERENT CENTROID SEPARATION METRICS ON SIMULATED HYPERSPECTRAL DATA WITH DIFFERENT MEAN TARGET PROPORTIONS. AVERAGE AUCS ARE SHOWN IN THE LAST COLUMN. BEST RESULTS ARE SHOWN IN BOLD, AND SECOND BEST RESULTS ARE UNDERLINED

Centroid Sep. Metrics	$\bar{p}_t$			Avg. AUC
	0.1	0.2	0.3	
Cosine	**0.786**	**0.857**	**0.882**	**0.841**
Euclidean	0.780	<u>0.855</u>	<u>0.874</u>	<u>0.836</u>
Manhattan	<u>0.782</u>	0.849	0.873	0.835

feature distribution with independent centroids and has fewer ambiguous or misplaced particles. We attribute the favorable properties of our method mainly to the metric-based regularizer, which can better reduce the intraclass variations and increase interclass separations than comparison methods. The feature distribution of max-attention MINN is inadequately differentiated compared with the other three methods in Fig. 7. It implies that only focusing on the samples with the highest attention scores and discarding the majority of instances is not always the optimal option. Fig. 7 also demonstrates that the proposed method can effectively extract the hidden target information from imprecisely labeled bags, contributing to the downstream target detection task.

4) Effect of Centroid Initialization Methods: We also conducted a comparative experiment to analyze the effect of different centroid initialization methods on the proposed MIML-Net. Three initialization methods, such as SND initialization, *Xavier* initialization [56], and *He* initialization [57], were included to provide comparisons. The corresponding performance results of MIML-Net using three different initialization methods on simulated hyperspectral data with varied mean target proportions are reported in Table III. It can be found in Table III that MIML-Net with these three initialization methods achieved similar detection performance, which verifies that the initialization method does not significantly affect the performance of the proposed MIML-Net. Also, all three variants yielded better results than the compared weakly supervised algorithms. Considering that SND initialization achieved an overall leading performance than the other two initialization methods, SND could be a more suitable choice for initializing centroids of MIML-Net.

5) Performance Analysis Using Different Centroid Separation Metrics: To validate the superiority of the centroid separation term implemented by cosine similarity ($\mathcal{L}_s = (1/\pi)\arccos \hat{\boldsymbol{\sigma}}_t^{\mathrm{T}} \hat{\boldsymbol{\sigma}}_b$) in the proposed MIML-Net, we compared it with two other similarity metrics on the simulated datasets for target detection. Euclidean distance ($\mathcal{L}_s^{\ell_2} = \|\boldsymbol{\sigma}_t - \boldsymbol{\sigma}_b\|_2$) and Manhattan distance ($\mathcal{L}_s^{\ell_1} = \|\boldsymbol{\sigma}_t - \boldsymbol{\sigma}_b\|_1$) were chosen for comparison. The performance results are shown in Table IV. It can be observed that cosine similarity excels in all three simulated data when compared with the other two metrics. We argue that the centroid separation via cosine similarity regularizes the feature space from the perspective of angle information, which is complementary to the magnitude information of embedding similarity term $\mathcal{L}_c$. Centroid separation achieved by Euclidean or Manhattan distances may not introduce additional angle information. Thus, cosine similarity could be appropriate for separating the target and background centroids. Moreover, MIML-Net with different types of centroid separation terms all provides performance benefits, indicating that centroid separation operation could enhance interclass dissimilarity and improve the discriminative capability of the detector.

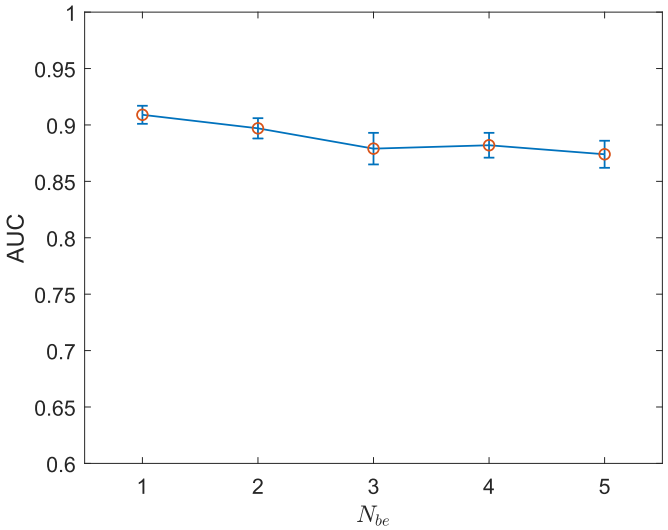

Fig. 8. Detection performance (mean AUC values and standard deviations) of MIML-Net on simulated data with varied numbers of background end-members N_{be}.

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TABLE V
LIST OF CONSTITUENT END-MEMBERS FOR BACKGROUND DIVERSITY ANALYSIS

Dataset Index	Target Endmember	Background Endmember
1	Red Slate	Pyroxenite
2	Red Slate	Pyroxenite, Verde Antique
3	Red Slate	Pyroxenite, Verde Antique, Phyllite
4	Red Slate	Pyroxenite, Verde Antique, Phyllite, Quartz Conglomerate
5	Red Slate	Pyroxenite, Verde Antique, Phyllite, Quartz Conglomerate, Basalt

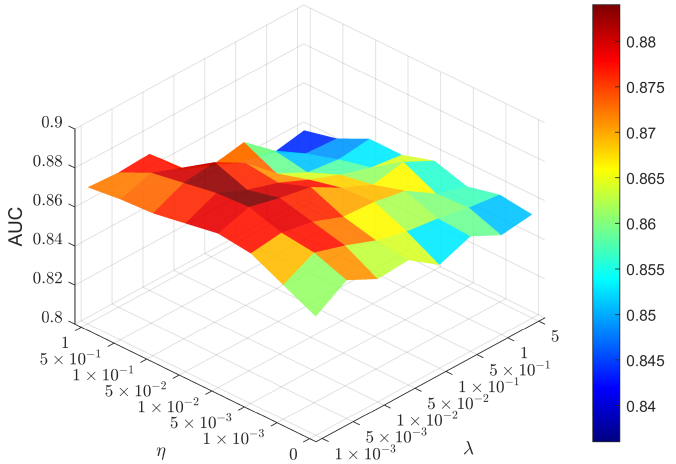

Fig. 9. Detection performance (AUC values) of grid search experiments on simulated data with mean target proportion $\bar{p}_t = 0.3$.

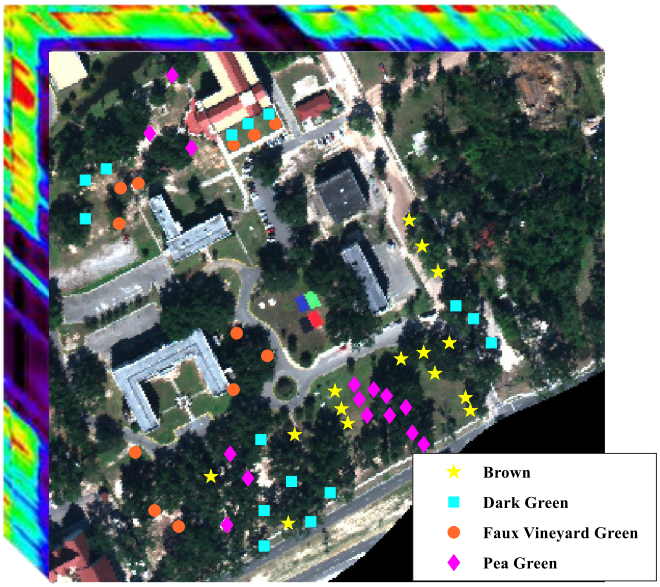

Fig. 10. One flight in the MUUFL Gulfport Hyperspectral Dataset and four kinds of scattered targets.

6) Effect of Background Diversity: The target detection task is challenging due to complex and diverse backgrounds [58]. The proposed algorithm dynamically regularizes the deep features of the background to one learnable cluster, which can facilitate the reduction of intraclass variations. To provide more profound insights into the impact of background diversity, we evaluated the proposed method on five simulated datasets with varied numbers of background end-members N_{be} ranging from 1 to 5. The mean target proportion $\bar{p}_t$ is set to 0.3. Gaussian white noise is added to control the SNR to 30 dB. The constituent end-members are listed in Table V. The mean AUC values and standard deviations of MIML-Net over five runs are presented in Fig. 8. As expected, the mean AUC values decrease, and the standard deviations show an increasing trend with the increase in N_{be}. However, the performance tends to be stable, as the background complexity continuously increases. It can be concluded that the MIML-Net is effective in background modeling and robust to the background diversity.

7) Effect of Parameters: We have conducted parameter sensitivity experiments on the simulated data with mean target proportion $\bar{p}_t = 0.3$ in a grid search manner to analyze the effect of parameter settings. The testing ranges of grid search experiments are shown in Table VI, and the detection performance is shown in Fig. 9, where the x-axis and the y-axis represent λ and η, respectively. In general, it can be seen in Fig. 9 that our method could achieve stable and satisfactory performance in a wide parameter range. Specifically, we can observe that the effectiveness of MIML-Net degrades when $\lambda > 1$. It may be because the excessive reduction of intraclass embedding similarity will affect the generalization ability of the learned features. Thus, $[1 \times 10^{-3}, 5 \times 10^{-2}]$ would be a reasonable choice for λ. Fixing the value range of λ, our algorithm tends to achieve higher AUC values when η takes a relatively greater value than λ, and there also exists a suitable interval for η, e.g., $[5 \times 10^{-3}, 1 \times 10^{-1}]$.

To further evaluate the robustness of MIML-Net to its parameters, we also performed a parameter perturbation analysis within the recommended parameter intervals ($\lambda \in [1 \times 10^{-3}, 5 \times 10^{-2}]$ and $\eta \in [5 \times 10^{-3}, 1 \times 10^{-1}]$). The best and worst performances (mean AUC over five runs) within the recommended ranges are 0.884 and 0.870, respectively. The percentage difference between the best and worst performance is only 1.6%, verifying the stability of the proposed method in a wide parameter range.

C. MUUFL Gulfport Hyperspectral Dataset

We further evaluate our method on the MUUFL Gulfport hyperspectral Dataset [59],[4] collected over The University of Southern Mississippi Long Beach on November 8, 2010. There are five collection flights over the site. The approximate spatial range of each scene is 330 × 340 m with a spatial resolution of 1 pixel/m^2. Each spectrum consists of 72 spectral bands corresponding to wavelengths of 375–1050 nm with a

[4]https://github.com/GatorSense/MUUFLGulfport

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TABLE VI
TESTING RANGES OF GRID SEARCH EXPERIMENTS FOR MIML-NET

Parameters	Testing Ranges
λ	1×10^{-3}, 5×10^{-3}, 1×10^{-2}, 5×10^{-2}, 1×10^{-1}, 5×10^{-1}, 1, 5
η	0, 1×10^{-3}, 5×10^{-3}, 1×10^{-2}, 5×10^{-2}, 1×10^{-1}, 5×10^{-1}, 1

TABLE VII
DETAILS OF EMPLACED WEAK TARGETS WITH VARIED SIZES IN THE MUUFL GULFPORT DATASET

Target Type	Num. of Targets with Different Sizes			Total Num.
	$0.5m\times0.5m$	$1m\times1m$	$3m\times3m$	
Brown	5	5	5	15
Dark Green	5	5	5	15
Faux Vineyard Green	4	4	4	12
Pea Green	5	5	5	15

10-nm sampling interval. Over the entire scenario, there are 57 manually placed cloth panels in four colors as targets, specifically Brown, Dark Green, Faux Vineyard Green, and Pea Green. The false color image of the hyperspectral scenario and the location of scattered targets are displayed in Fig. 10. The numbers of placement targets with different sizes are illustrated in Table VII. This open-field dataset was captured under varied light and occlusion conditions due to the shadow and tree coverage, creating a challenging target detection problem. It is infeasible to manually identify the location of the targets by visual examination. The ground-truth location of targets is collected according to the GPS information measured by both airborne and ground-based instruments. Alignment errors may result in a 1–2-m shift between the target marker and its true position.

1) Single-Class Hyperspectral Target Detection: In this experimental configuration, we choose Flights 1–3 as training images, Flight 4 for validation, and Flight 5 for testing. We generate MIL data bags with the help of GPS information. Specifically, a 5×5 neighborhood area anchored by the GPS coordinate of each target is aggregated into a positive bag. The rest area without any type of target is divided into multiple negative bags. The size of each negative bag is set to 25, which is the same as the size of the positive bag to facilitate parallel computation in the training process. The following comparison experiments were conducted on these data bags with binary labels.

The normalized area under the ROC curves (NAUCs)[5] at FPR $= 1 \times 10^{-3}$ is employed for performance evaluation [12]. Table VIII provides the NAUC indicators of MIML-Net and comparison algorithms. Our method always obtained the highest or the second NAUC values on four types of targets. MIML-Net also achieved the highest average NAUC, verifying its overall effectiveness on target detection. Compared with the second and third best algorithms from the view of average NAUCs, $L1$-attention MINN, and MI-SMF, our method yields overall performance improvements of 4.3% and 17.6%, respectively. The ROC curves for detected Dark Green targets at 1×10^{-3} FPR are shown in Fig. 11,

[5]https://github.com/GatorSense/MUUFLGulfport/tree/master/MUUFLGulfportDataCollection/Bullwinkle

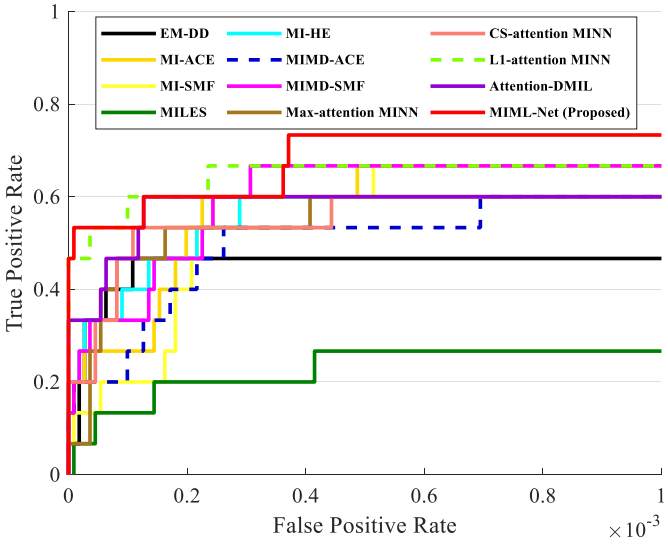

Fig. 11. ROC curves of MIML-Net and comparison algorithms on the Gulfport dataset for Dark Green target detection task.

where the ROC curve of MIML-Net is generally higher than other comparison algorithms, especially at $[0.4, 1] \times 10^{-3}$ FPR range, proving the advancedness of our algorithm.

2) Multiclass Hyperspectral Target Detection: In this setting, we included a multiclass target detection task to further verify the effectiveness of the proposed MIML-Net, where four types of targets (Brown, Dark Green, Faux Vineyard Green, and Pea Green) are joined together. For each HSI, there are 57 positive bags. Similarly, the negative bags are created by the remaining area without any type of targets. Flights 1–3 serve as the training set, and Flights 4 and 5 serve as the validation and test sets, respectively.

Table IX and Fig. 12 report the NAUC values and ROC curves of MIML-Net and comparison techniques. The ROC curve of MIML-Net is remarkably higher than most comparison techniques. From the perspective of NAUC values, MIML-Net also yields competitive performance.

D. HyMap Target Detection Campaign

Except for the Gulfport dataset, we have included another comparative experiment, the HyMap target detection campaign [60], to further verify the effectiveness of our proposed algorithm. The HyMap dataset was collected in July 2006 over Cook City, MT, USA. Each image contains 280×800 pixels with 3-m ground resolution, and 126 spectral bands that correspond to wavelengths ranging from 450 to 2500 nm with approximately 15-nm bandwidth. Four types of fabric panels (F1–F4) and three types of vehicles (V1–V3) are emplaced as targets of interest. Fig. 13 presents the ground-truth locations of targets, and Table X lists the detailed information. Two

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TABLE VIII

NAUCs AT FAR $= 1 \times 10^{-3}$ FOR MIML-NET AND COMPARISON ALGORITHMS ON THE GULFPORT DATASET WITH INDIVIDUAL TARGET TYPE DETECTION. AVERAGE NAUCs ARE SHOWN IN THE LAST COLUMN. BEST RESULTS ARE SHOWN IN BOLD, AND SECOND BEST RESULTS ARE UNDERLINED

Alg.	Target type				Avg. NAUC
	Brown	Dark Green	Faux Vineyard Green	Pea Green	
EM-DD [31]	0.453	0.449	0.207	0.042	0.288
MI-ACE [12]	0.521	0.572	0.397	0.285	0.444
MI-SMF [12]	0.520	0.549	<u>0.481</u>	0.267	0.454
MILES [33]	0.199	0.226	0.275	0.077	0.194
MI-HE [15]	0.510	0.548	0.333	0.302	0.423
MIMD-ACE [16]	0.520	0.495	0.159	0.298	0.368
MIMD-SMF [16]	0.497	0.592	0.420	0.247	0.439
Max-attention MINN [53]	0.580	0.542	0.227	0.302	0.413
CS-attention MINN [35]	0.498	0.546	0.326	0.317	0.422
L1-attention MINN [22]	**0.617**	<u>0.642</u>	0.437	<u>0.355</u>	<u>0.512</u>
Attention-DMIL [21]	0.440	0.576	0.347	0.283	0.412
MIML-Net (proposed)	<u>0.587</u>	**0.675**	**0.503**	**0.370**	**0.534**

TABLE IX

NAUCs AT FAR $= 1 \times 10^{-3}$ FOR MIML-NET AND COMPARISON ALGORITHMS ON THE GULFPORT DATASET WITH ALL FOUR TARGET TYPES DETECTION. BEST RESULTS ARE SHOWN IN BOLD, AND SECOND BEST RESULTS ARE UNDERLINED

Alg.	NAUC	Alg.	NAUC
EM-DD [31]	0.256	MIMD-SMF [16]	0.272
MI-ACE [12]	0.282	Max-attention MINN [53]	0.166
MI-SMF [12]	0.226	CS-attention MINN [35]	0.266
MILES [33]	0.162	L1-attention MINN [22]	**0.433**
MI-HE [15]	0.279	Attention-DMIL [21]	0.315
MIMD-ACE [16]	0.246	**MIML-Net** (proposed)	<u>0.423</u>

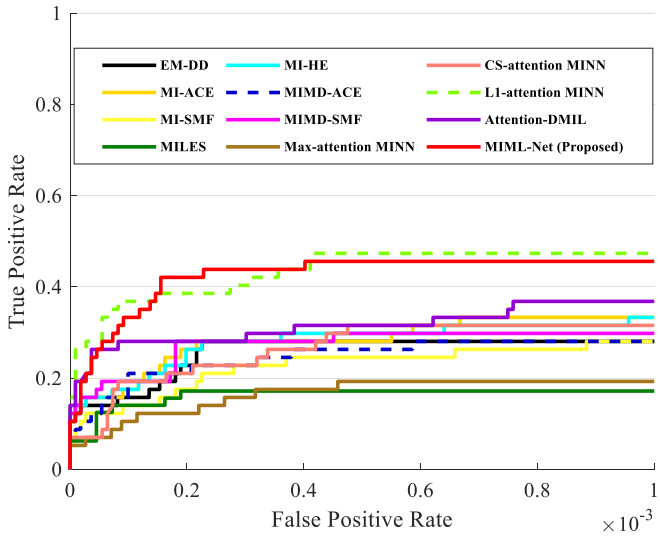

Fig. 12. ROC curves of MIML-Net and comparison algorithms on the Gulfport dataset for multiclass target detection task.

HSIs of the same scene, such as a self-test image (STI) and a blind-test image (BTI), were used for validation. STI has pixel-level target ground-truth locations, while BTI is unlabeled. For BTI, we manually segment a 5 × 5 area around each target as a positive bag by referring to the location of the targets in STI. The remaining regions without any targets are separated into multiple negative bags. A multiclass hyperspectral target detection has now been created. The data bags from BTI with bag-level labels are used as training data. The STI with pixel-level labels is served as testing data.

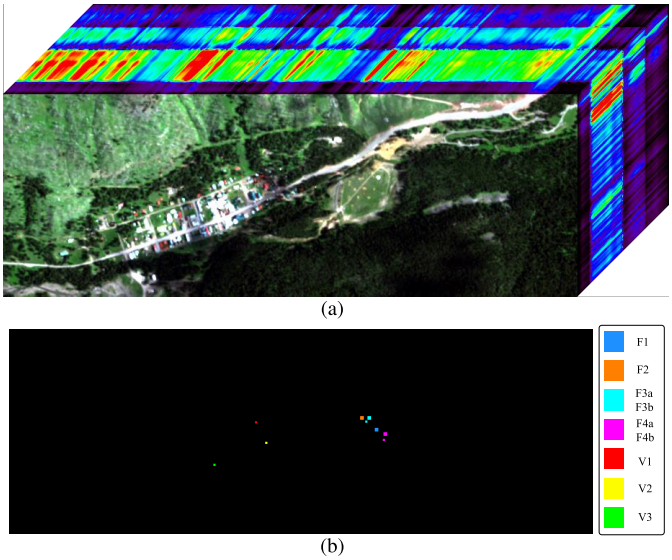

Fig. 13. (a) HyMap hyperspectral dataset RGB image. (b) Seven types of scattered targets.

TABLE X

INFORMATION OF SCATTERED TARGETS IN HYMAP DATA

Target Index	Type	Size	Pixels
F1	Red cotton	$3m \times 3m$	9
F2	Yellow nylon	$3m \times 3m$	9
F3a	Blue cotton	$2m \times 2m$	9
F3b	Blue cotton	$1m \times 1m$	1
F4a	Red nylon	$2m \times 2m$	9
F4b	Red nylon	$1m \times 1m$	1
V1	Chevy Blazer	N.A.	1
V2	Toyota T100	N.A.	1
V3	Subaru GL Wagon	N.A.	1

Fig. 14 visualizes the ROC curves of all comparison methods, where the ROC curve of MIML-Net is evidently higher than the state-of-the-art comparisons, especially in [0.2, 1] false alarm rate (FAR) range. Table XI provides the corresponding AUC values for quantitative evaluation. MIML-Net obtains the highest AUC performance, verifying its overall effectiveness on target detection.

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TABLE XI

AREA UNDER THE ROC CURVES (AUCS) FOR MIML-NET AND COMPARISON ALGORITHMS ON HYMAP DATA WITH ALL SEVEN TARGET TYPES DETECTION. BEST RESULTS ARE SHOWN IN BOLD, AND SECOND BEST RESULTS ARE UNDERLINED

Alg.	AUC	Alg.	AUC
EM-DD [31]	0.934	MIMD-SMF [16]	0.958
MI-ACE [12]	0.960	Max-attention MINN [53]	0.965
MI-SMF [12]	0.944	CS-attention MINN [35]	0.953
MILES [33]	0.902	L1-attention MINN [22]	<u>0.971</u>
MI-HE [15]	0.945	Attention-DMIL [21]	0.964
MIMD-ACE [16]	0.953	**MIML-Net** (proposed)	**0.984**

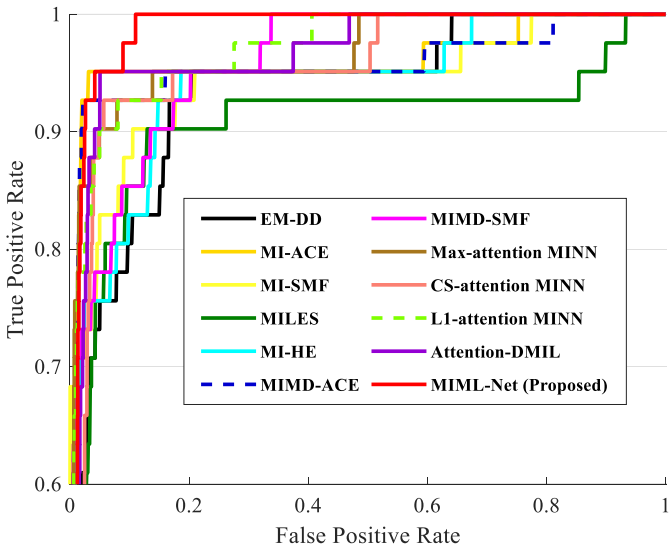

Fig. 14. ROCs for the proposed MIML-Net and comparison algorithms on the HyMap dataset.

V. CONCLUSION

In this article, we proposed a novel weakly supervised method, MIML-Net, for hyperspectral target detection with ambiguously labeled data. Our method aggregates hidden target information under MIL attention-based framework and further constrains target and background embeddings to two learnable compact clusters with separate centroids, encouraging discriminative feature learning. We perform an extensive empirical evaluation over simulated and real-field datasets with lots of weakly supervised techniques, which demonstrates that MIML-Net is more effective. As for future potential research directions related to this work, we may focus on exploring the semantic information within bags and modeling the correlation among instances.

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